

A Method for the Automatic Generation of Technical Manuals for Civil Aircraft Wiring Lists

Shuang Zhang

Shanghai Aircraft Customer Service Co. Ltd, Commercial Aircraft Corporation of China Ltd.

SACSC, COMAC

Shanghai, China

13761962689@163.com

Abstract—Wiring manuals contain the complex and extensive information of an entire aircraft's wiring and electrical equipment. These manuals are characterized by large data volumes and frequent updates. Historically, domestic civil aircraft manufacturers have primarily relied on manual methods to compile wiring list technical manuals. Experience has shown that this approach is prone to inefficiencies in compilation and inaccuracies in content. Aircraft wiring list technical manuals are not only critical references for troubleshooting wiring issues during operations but also serve as essential inputs for replacing and maintaining electrical standard components. This study proposes a method for the automatic generation of civil aircraft wiring list technical manuals by studying the evaluation rules for technical documents, methods for extracting and reconstructing technical elements of wiring lists, and the data structures of wiring list technical elements under the S1000D standard. The method incorporates the requirements of civil aviation continued airworthiness documentation and is verified for feasibility through its application in the compilation of model-specific manuals. The research results demonstrate that the proposed method for automatically generating aircraft wiring list technical manuals is feasible. It significantly improves the efficiency of manual compilation and the accuracy of content, showing great potential for application in the automatic generation of wiring list manuals by aircraft manufacturers. (*Abstract*)

Keywords—Civil Aircraft; Parsing; Data Reconstruction; S1000D; Automatic Generation; Wiring Manual (*key words*)

I. INTRODUCTION

According to the requirements for continued airworthiness maintenance data in the Civil Aviation Convention [1-2], aircraft manufacturers must provide operators with foundational information about wiring, electrical components, and electrical component connections that match the actual physical state of the aircraft. These pieces of information serve as critical technical elements for troubleshooting wiring issues (e.g., connector part numbers in the electrical component information group are used as a basis for replacing faulty plugs and sockets during operations, while termination codes in the wiring connection details guide crimping processes for terminal pins). These technical elements are typically compiled by aircraft manufacturers into wiring list manuals for use by operators, ensuring the continued safe operation of the aircraft.

International civil aircraft manufacturers have achieved a high level of integration in their manuals. With the promotion of interactive electronic technical manuals for civil aircraft,

which incorporate interactive functions between diagrams and manuals [3-5], higher demands have been placed on the structural organization of technical manual data [6]. The wiring list manuals, which carry information on wires and electrical components, often consist of hundreds of thousands of entries. The structured storage of data in these manuals is a necessary condition for enabling interaction between wiring diagrams and maintenance procedures [7]. In civil aviation, general aviation, maritime, and other fields, the S1000D standard [8] has been widely applied in the development of structured manuals [9-13]. This study explores the automatic generation of wiring lists based on the widely adopted S1000D standard.

The initial drafting of an aircraft's wiring list manual typically derives from wiring design data. In the early stages of the civil aviation industry, aircraft wiring design data were often published by manufacturers in the form of low-structured 2D drawings or PDFs within their design documents. The process of extracting design data, conducting configuration comparisons, and inputting the data into S1000D-compliant manual modules relied heavily on extensive manual effort. With the increasing number of aircraft produced by major manufacturers, the optimization of design functionality, and the delivery of aircraft to customers, the volume of wiring list data has grown exponentially. This traditional manual process is inefficient and prone to errors.

In recent years, with the development of the civil aviation industry, aircraft design data have gradually transitioned toward greater structural organization, providing favorable conditions for the automatic generation of wiring list manuals. However, existing research on automatic generation of wiring lists relies heavily on CHS or other drawing software [14-15]. Based on practical experience, the limitations of those drawing software—particularly its lack of configuration management mechanisms for electrical components within its property pages—restrict this method to the automatic generation of wiring lists for individual aircraft. This approach fails to meet the requirements of manufacturers needing to generate manuals for multiple aircraft at scale.

This study proposes a method for the automatic generation of wiring list manuals for both single and multiple aircraft based on the S1000D standard. This method does not depend on mature commercial software but instead relies on a self-developed tool. Using the wiring list manual of a specific aircraft model in airline operation as an example, the feasibility and practical value of this method are demonstrated.

II. OVERALL DESIGN

The overall design model for the automatic generation method of aircraft wiring list manuals based on the S1000D standard is shown in Figure 1.

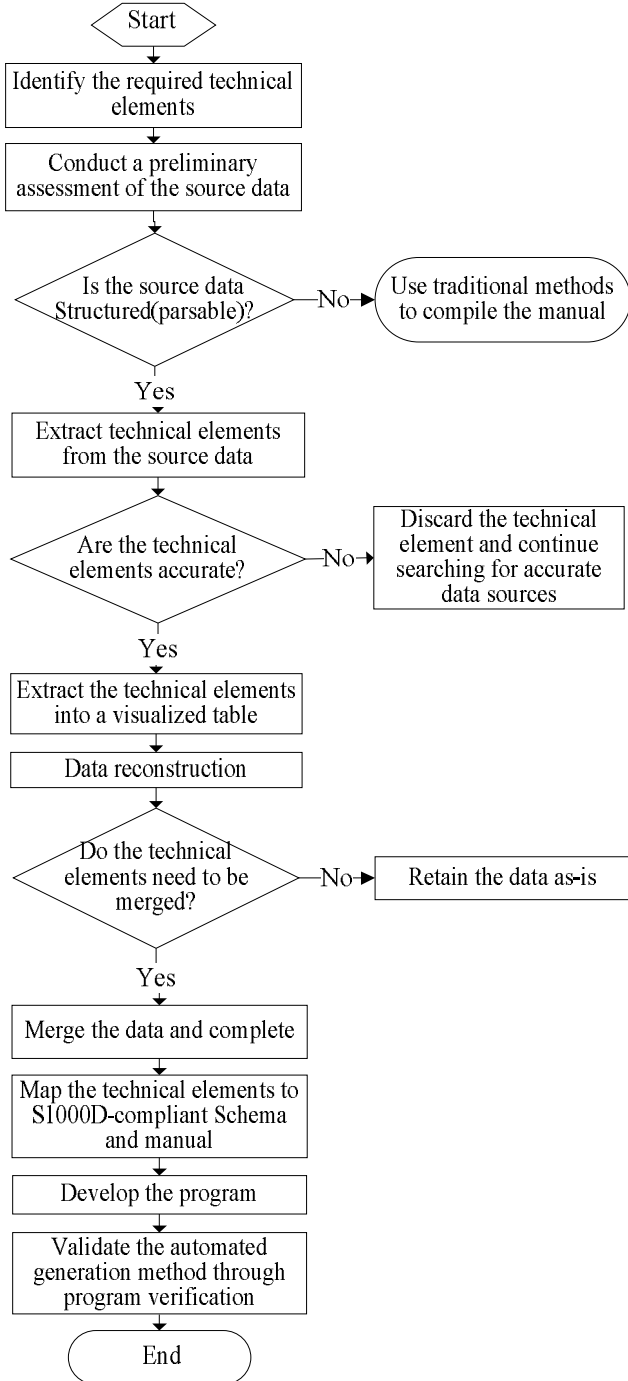


Figure 1. Overall Design Model for the Automatic Compilation of Wiring List Manuals

First, the technical elements required for the wiring list manual must be identified. Before automatically generating manual data modules, it is necessary to identify the technical elements to be provided to the customer in the manual by

taking into account airworthiness regulations, industry standards for civil aviation publications, and customer requirements.

Second, the process logic for automatically obtaining technical elements from source data and automatically generating manual list data modules is developed. This process is divided into five substeps: initial assessment of source data, analysis and extraction of technical elements, data reconstruction, matching with manual data modules and schema, and program development. In this process, the types of source data suitable for extraction via automated programs should be analyzed to serve as the research objects. The following must be explored: the logic for extracting technical elements from the parsed source data, the logic for reconstructing data based on the extracted technical elements, and the logic for matching reconstructed data with manual data modules. Based on these logical approaches, a program for automatically generating manual list data modules is developed.

Finally, the application of the automatic generation method is verified through case studies using specific aircraft models.

III. MODULE DESIGN

A. Identifying the Technical Elements Required for the Wiring List Manual

During the initial stage of compiling the wiring list manual, it is necessary to identify the technical elements that the data modules in such publications should include. By interpreting the requirements for continued airworthiness data in the International Civil Aviation Convention, analyzing the internationally recognized S1000D publication standards, and collecting the wiring list requirements of airlines during route operations, the technical elements required to meet the operational and maintenance needs of wiring list manuals have been summarized. The technical elements required for route maintenance in the wiring list manual are detailed in Table I.

TABLE I. TECHNICAL ELEMENTS REQUIRED FOR WIRING LIST MANUALS

Element	Sub-Element	Description
Wire Information Group	Wire Number	Wire Unique Identification Code
	Wire Material	Wire Material Code
	Wire Gauge	Wire Diameter Specification Code
	Wire Length	Wire Length (mm/feet)
	From/To Terminal Equipment Number	Equipment Codes Connected at Both Ends of the Wire
	From/To Terminal Equipment Pin	Pins of the Equipment Connected at Both Ends of the Wire
	From/To Terminal Termination Code	Program Code for Terminal Termination Procedures on Both Ends of the Wire
	Routing Path	Code Related to Wire Routing and Signal Functions
	Associated Wiring Diagram Number	Code for the Wiring Diagram the Wire Belongs To
	Applicability	Information About the Aircraft Serial Numbers Applicable to the Wire

Element	Sub-Element	Description
Electrical Component Information Group	Equipment Number	Equipment Unique Identification Code
	Equipment Type	Connector/Relay/Circuit Breaker/Wiring Module/Grounding Point/Grounding Module/Permanent Joint
	Standard Part Number	Standard Part Number Used for the Installed Equipment
	Location Code	Installation Location Code of the Equipment on the Aircraft
	Associated Wiring Diagram Number	Code for the Wiring Diagram the Equipment Belongs To
	Applicability	Information About the Aircraft Serial Numbers Applicable to the Equipment

B. Initial Assessment of Source Data

The first step in the automatic generation of the wiring list manual is the initial assessment of the source data to confirm the types of source data that can be parsed. To ensure consistency with the actual physical configuration of the aircraft, the wiring list manual should evaluate data originating from the engineering design phase, manufacturing phase, and customer operation phase. The parsability of the source data is a prerequisite for the automatic generation of publications. It is first necessary to determine whether the updates or revisions of the source data are structured and capable of being extracted in bulk. The feasibility analysis of batch extraction from source data is summarized in Table II.

TABLE II. FEASIBILITY ANALYSIS OF BATCH EXTRACTION FOR WIRING LIST SOURCE DATA

Source	Source Data Type	File Structure	Parsable (Y/N)
Engineering Design Phase	Harness Installation Digital Model	XML/ EXCEL	Y
	Structured Wiring Diagram	XML/ EXCEL	Y
	Equipment Installation Digital Model	XML/ EXCEL	Y
	Structured Forms	EXCEL	Y
Manufacturing Phase	Concession Reports	PDF/ WORD	N
Operations Phase	Service Bulletins	PDF/ WORD	N

From Table II, it can be seen that the source data from the engineering phase meets the necessary conditions for parsability. The process of using engineering source data to generate the wiring list manual can be automated. Considering that 90% of the data used in compiling the aircraft wiring list manual originates from the engineering phase, it is determined that using engineering source data as the research object holds significant research value.

C. Analysis and Extraction of Technical Elements

The second step in the automatic generation of the wiring list manual involves analyzing source data from the engineering phase to extract technical elements. Due to time discrepancies in the data transfer processes among various engineering research departments, inconsistencies may exist for the same technical elements across different engineering data sources. To ensure that the data included in the wiring list manual is consistent with the physical configuration of the aircraft, rules for determining the data sources of each technical element must be established before analysis.

Using the wire and equipment information groups as research objects, the logic included in the automated processing program for determining the data sources of their technical elements is as follows:

- If the wire or equipment is a finished product provided by a supplier, the basic technical elements of the wire (e.g., wire material, gauge, length) and the technical elements of the equipment should be extracted from the data source of the corresponding professional research department of the supplier, as this is the precise data source; the wiring connection technical elements (e.g., “from/to” terminal equipment, pin positions, termination codes, routing paths etc.) should be extracted from the data source of the aircraft manufacturer’s wiring layout research department, as this is the precise data source.
- If the wire or equipment is a self-manufactured product of the main manufacturer, the technical elements of the wire and equipment should be extracted from the data source of the aircraft manufacturer’s wiring layout research department, as this is the precise data source.

Taking a finished wire from the water and wastewater system and an electrical component manufactured by the main manufacturer as examples, some of the determination examples are shown in Table III.

TABLE III. EXAMPLES OF DETERMINING TECHNICAL ELEMENT EXTRACTION FROM ENGINEERING DATA

Element	Sub-Element	Example	Source Data Type	Source Data Responsible Party	Data Accuracy (Y/N)	Extraction Determined (Y/N)
Wire Information Group (Finished Wire)	Wire Material	/	Structured Wiring Diagram	Design EWIS Specialty	N	N
		M27500-20SE2U00	Structured Finished Wire Form	Design Water/Wastewater Specialty	Y	Y
	Wire Gauge	/	Structured Wiring Diagram	Design EWIS Specialty	N	N
		22	Structured Finished Wire Form	Design Water/Wastewater	Y	Y

Element	Sub-Element	Example	Source Data Type	Source Data Responsible Party	Data Accuracy (Y/N)	Extraction Determined (Y/N)
	From/To Terminal Equipment Number	L-33J1-P/D-52B-P	Structured Wiring Diagram	Specialty Design EWIS Specialty	Y	Y
		/	Structured Finished Wire Form	Design Water/Wastewater Specialty	N	N
	From/To Terminal Equipment Pin	A/J	Structured Wiring Diagram	Design EWIS Specialty	Y	Y
		/	Structured Finished Wire Form	Design Water/Wastewater Specialty	N	N
	From/To Terminal Termination Code	519/520	Structured Wiring Diagram	Design EWIS Specialty	Y	Y
		/	Structured Finished Wire Form	Design Water/Wastewater Specialty	N	N
Electrical Component Information Group (Main Manufacturer Part)	Equipment Type	Connector	Harness Installation Digital Model	Design EWIS Specialty	Y	Y
	Standard Part Number	M39012/26-0011	Harness Installation Digital Model	Design EWIS Specialty	Y	Y
	Location Code	128 ST10581	Harness Installation Digital Model	Design EWIS Specialty	Y	Y

For technical elements where the extraction result is determined to be “Y”, the extraction should be performed from structured engineering data. Typically, the storage rules for technical elements in structured data are traceable. First, the structure of the engineering data should be analyzed to understand the storage logic and rules for each technical element, preparing for the batch extraction of these elements. Next, the extracted technical elements should be visualized and stored in an Excel table, providing the data foundation for the third step, data reconstruction.

Using the engineering data for harness installation of a specific aircraft model as an example, the logic for extracting the technical elements “Wire Length” and “From/To Terminal Equipment Numbers” is as follows:

- 1) Determine whether the value of the “Type” attribute in the ‘<WireList/Wire>’ node is “Normal”.
 - a) If it is “Normal”, retrieve the value of the “Length” attribute in the ‘<WireList/Wire>’ node as the “Wire Length” technical element and save it to the Excel table, e.g., “256”.
 - b) If not, proceed to the next step.
- 2) Determine whether the value of the “Type” attribute in the ‘<WireList/WireGroup/Wire>’ node is “Shield”.
 - a) If it is “Shield”, the wire is identified as a shielded wire, and the “Wire Length” technical element is left blank and saved to the Excel table.

- 3) Determine whether there are two ‘<WireExtremity>’ nodes under ‘<WiresExtremitiesList>’ with the same “WireTag” attribute value.

a) If such nodes exist, take the value of the “PinParent” attribute from the node with the smaller “ID” value as the “From Terminal Equipment Number” and the value of the “PinParent” attribute from the node with the larger “ID” value as the “To Terminal Equipment Number”. Save these technical elements to the Excel table, e.g., “D13511-A1P”.

b) If no such nodes exist, continue searching for the next pair of nodes with identical “WireTag” attribute values.

- 4) Iterate through all sections and chapters of the engineering data for the entire aircraft, ensuring that all wires and equipment are parsed and extracted into the table.

Following the example above, similar extraction logic can be applied at the granularity of individual engineering data files. For instance, using an engineering data file of a large passenger aircraft, the program can batch extract partial equipment technical elements from the electrical component information group. Examples of these extractions are shown in Table IV. The extraction logic for the technical elements of the wire information group follows a similar approach and will not be repeated here.

TABLE IV. EXAMPLES OF EXTRACTING TECHNICAL ELEMENTS OF ELECTRICAL COMPONENT INFORMATION GROUP FROM ENGINEERING DATA

Equipment Number	Equipment Type	Equipment Part Number	Location	Diagram Number	Data Source File Number
D175-AP	Connector	D389/MA3SN	ST615	32-32	10C7100G20
W1602-SM1	Permanent Joint	CETS0004-0177	ST619	33-13	10C7100G20
M1112-L	Wiring Module	M81714/60-22-02	ST678	26-21	10C7100G20

D. Data Reconstruction

The third step in the automatic generation of the wiring list manual is to reconstruct the technical elements according to configuration rules. Each engineering data file inherently contains properties indicating the applicable aircraft serial numbers. However, technical elements extracted from different engineering data files often have overlapping information, meaning that the content of certain technical elements is identical across different serial numbers. According to the recommendations of the S1000D standard, identical technical element content should be merged based on configuration, reconstructing the originally serial-number-specific storage into configuration-specific storage. To avoid redundant information in the wiring list manual and to ensure concise and accurate technical information for maintenance personnel, the technical elements must be reconstructed. The reconstructed data will serve as the input for the fourth step, which involves mapping the data to the S1000D-compliant manual data modules.

The automated processing program for data reconstruction should include the following logical elements:

- 1) *Wire Information Group*: Use the wire ID as the unique identifier. From the data extracted from different applicable serial numbers in the engineering files, find the corresponding technical elements for the wire and reconstruct them based on configuration.
- 2) *Electrical Component Information Group*: Use the equipment ID as the unique identifier. From the data extracted from different applicable serial numbers in the engineering files, find the corresponding technical elements for the equipment and reconstruct them based on configuration.

3) *Configuration Reconstruction Process*: First, remove the serial number-specific applicability information. Then compare each technical element item by item.

4) *Comparison Results*: For items with identical comparison results, apply configuration aggregation (e.g., adding applicability) and return the reconstructed configuration value. For items with different comparison results, retain the original validity values as independent configurations without modification.

5) *Full Aircraft Reconstruction*: Traverse the entire parsed table and perform configuration reconstruction at the granularity of wires and equipment.

6) *Marking Reconstructed Data*: Mark the data that has undergone configuration reconstruction in the engineering files.

Using the connector D174-AP from the electrical component information group of a large passenger aircraft as an example, the data before reconstruction is shown in items 1-6 of Table V. This data was extracted from the engineering files 10C7100G20 (applicable to serial number 00001) and 10C7100G21 (applicable to serial number 00002). A comparison of the technical elements extracted from these two sources shows that their content is completely consistent. As a result, the applicable serial numbers should be merged and stored as 00001-00002, meaning items 1-2 should be merged, and items 4-6 should also be merged. The reconstructed data is shown in items 7-9, thus completing the configuration reconstruction for the connector D174-AP. The reconstruction logic for the wire information group follows a similar process and will not be elaborated further.

TABLE V. EXAMPLES OF EXTRACTING TECHNICAL ELEMENTS OF THE WIRE INFORMATION GROUP FROM ENGINEERING DATA

Index	Equipment Number	Equipment Type	Equipment Part Number	Location	Associated Diagram Number	Data Source Number	Applicable Serial Number	Merge Required (Y/N)	Reconstructed (Y/N)
1	D175-AP	Connector	D389/MA3SN	ST615	32-32-0	10C7100G20	00001	Y	N
2	D175-AP	Connector	D389/MA3SN	ST615	32-32-0	10C7100G21	00002	Y	N
3	D175-AP	Connector	CECN9018-RBD-50-L-00	ST615	34-31-0	10C5100G20	00003	N	N
4	D175-AP	Connector	D389/MH6SN	ST615	44-10-0	10C5600G20	00004	Y	N
5	D175-AP	Connector	D389/MH6SN	ST615	44-10-0	10C5600G21	00005	Y	N
6	D175-AP	Connector	D389/MH6SN	ST615	44-10-0	10C5600G22	00006-00019	Y	N
7	D175-AP	Connector	D389/MA3SN	ST615	32-32-0	10C7100G20	00001-00002	/	Y
8	D175-AP	Connector	CECN9018-RBD-50-L-00	ST615	34-31-0	10C5100G20	00003	/	Y
9	D175-AP	Connector	D389/MH6SN	ST615	44-10-0	10C5600G20	00004-00019	/	Y

E. Matching Manual Data Modules and Schema

The wire information group and electrical component information group correspond to the wire list data module and

equipment list data module, respectively, in the wiring publication. The fourth step in the automatic generation of the wiring list manual involves matching standardized codes to generate the wire list and equipment list data modules.

As specified in S1000D [8], the information codes in data module coding are used to define the types of data modules. The 056 and 057 information codes can be used to define equipment lists and wiring lists. Different information codes correspond to different Schema categories to distinguish the storage formats of technical elements, with the wiringdata Schema defined for wiring data storage.

According to the S1000D standard: The 056 information code and the wireGroup subclass of the wiringdata.xsd Schema can match the data storage rules for the wire list data module. The 057 information code and the electricalEquipGroup subclass of the wiringdata.xsd Schema can match the data storage rules for the equipment list data module. The structure of the Schema is shown in Figure 2.

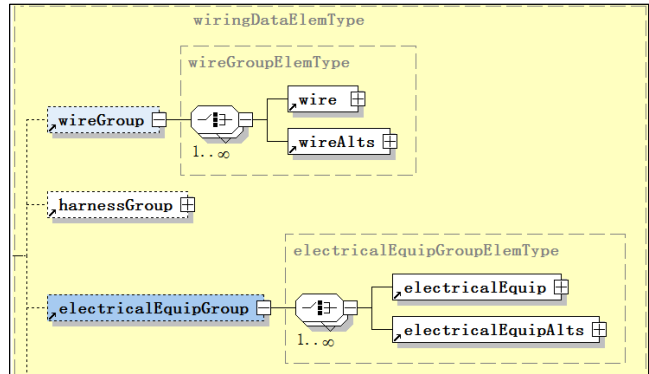


Figure 2. Schema for Wire and Equipment List Data Modules

By defining appropriate codes within the Schema subclasses, data storage redundancy can be reduced, resulting in a concise and clear data storage structure for the list. This approach facilitates subsequent data tagging and necessary extraction. The mapping relationships between the data modules of the wire information group and electrical component information group, their information codes, and the corresponding Schema are shown in Table VI.

TABLE VI. MAPPING RULES OF TECHNICAL ELEMENTS TO DATA MODULES

Mapping Schema	Mapping Module	Element
/@wireNumber	057A	Wire Number
/@wireType	057A	Wire Material
/@wireGauge	057A	Wire Gauge
/@length	057A	Wire Length
/fromEquip//@functionalItemNumber /toEquip//@functionalItemNumber	057 A	From/To Terminal Equipment Number
/fromEquip//@contactIdent /toEquip//@contactIdent	057A	From/To Terminal Pin
/fromEquip//@specialConnection /toEquip//@specialConnection	057A	From/To Terminal Termination Code
/@wireRoute	057A	Routing Code
/functionalItemRef/@functionalItemNumber	056A/ 057B~G	Equipment Number
/electricalEquip/@equipName	056A/ 057B~G	Equipment Name

Mapping Schema	Mapping Module	Element
/electricalEquip/@partNumber	056A/ 057B~G	Equipment Part Number
/@installationInfoInstallation Location	056A/ 057B~G	Installation Location
/dmCode/	056A/ 057B~G	Associated Diagram Number

When matching technical elements to data modules, the program for automatically generating data modules should include the following logical elements:

- 1) Map the technical elements of the wire information group to the 057A information code data module according to Table VI.
- 2) Map the technical elements of the equipment information group to the 056A/057B~G information codes according to Table VI, with the information code distinguished based on the equipment type. For example, connector-type equipment can use 057B.
- 3) The data module coding should correspond to the harness number, equipment type, and equipment number, with rules defined based on the characteristics of the aircraft model.
- 4) Store data modules using a WireAlts group for the same wire number and an EquipmentAlts group for the same equipment number.
- 5) In a customized data package, if a specific data module contains no content for the selected applicable serial numbers, that data module should be deleted.

F. Program Development

According to the logic outlined above, the rules in Sections 2.2-2.5 should be implemented in Python or equivalent programming code to automate the four-step process described in these sections, thereby developing a program tool for the automatic generation of the wiring list manual.

IV. APPLICATION RESULTS

By utilizing the program tool, the application of the automatic generation method for wiring list manuals based on the S1000D standard has been realized. Taking a large passenger aircraft as an example, the generated interactive manual includes examples of wire list data modules (as shown in Figure 3) and equipment list data modules (as shown in Figure 4).

Wire No.	Wire Code	Wire Ga.	Len. mm(in.)	From Equip.	Pin	Term.	To Equip.	Pin	Term.	Routing Code	Diagram No.
W1012-24001-16NBL	132	16	1263(50)	D1922-A1P	A	544	U-280-P2	B	544	3N-ESS-D-24-P03-C	24-23-11
W1012-24001-1	132	16	1263(50)	D1922-A1P	B	544	U-280-P2	A	544	3N-ESS-D-24-P03-C	24-23-11

Figure 3. Publication Effect of Wire List Data Module

Equipment No.	Description	Part Number	Location	Diagram
2D-215-P1	Connector	M24308/2-1F(M85049/48-2-1F)	122 STA5915.1	21-25-00
2D-271A-P1	Connector	D38999/26MC355A(CEBK0500HX13)	122 STA3961.8	27-92-00
Applicability: 0008-00016,00033				
2D-232-P1	Connector	CECN9024-TV06RF-9-95(CEBK0303-0804)	221 STA8708.5	23-75-00
Applicability: ALL				
2D-21P-P1	Connector	D38999/26FA355A(CEBK0103-9)	125 STA7242.4	21-25-00
2D-32C-P1	Connector	D38999/26KA985A(CEBK7003HZ1-0904)	124 STA8694.7	32-60-00

Wiring Equipment List-Connector-2D
C9A9-A-91-21-2D-COA-057D-A

Figure 4. Publication Effect of Equipment List Data Module

This method has been widely applied in the initial compilation and optimization processes of multi-serial-number wiring list manuals. It eliminates the time-consuming and labor-intensive nature of large-scale optimizations such as restructuring the manual and splitting data modules. Moreover, the generated S1000D-compliant data modules provide robust data support for interactions between diagrams and lists, as well as interactions with other manuals.

V. CONCLUSION

The aircraft wiring list manual automatic generation method based on the S1000D standard proposed in this study is a feasible model for the automatic generation of structured manual data from engineering wiring data, with a clearly defined implementation pathway.

For technical optimization, this method eliminates the risks of errors in manual compilation caused by human factors. Compared to the drawing software-based method for generating wiring list technical manuals, it resolves limitations stemming from configuration control issues that restrict the generation of manuals to a single aircraft instance, enabling the automatic compilation of wiring list technical manuals for multiple aircraft.

For practical application, based on three years of operational experience with a large passenger aircraft, this method saved 16,300 labor hours during the initial manual compilation stage compared to traditional manual methods. Additionally, each quarterly manual revision process achieved an average time savings of 2,800 labor hours, improving compilation efficiency by 95.8%. Manuals compiled using this method resulted in an average annual reduction of 36 customer-reported issues and a 97.2% reduction in content error rates. The method has been applied in 12 major manufacturer editions and customized manual compilations for the aircraft model. Each manual revision using this method has led to the publication of over 30,000 pages of manual content.

For future development, beyond the civil aviation field, this method can also be applied to other industries with complex wiring systems. During practical applications, it is essential to analyze the hierarchical structure of structured wiring data sources and appropriately adjust the technical elements according to the needs of manual users. Customization should

be carried out to address domain-specific differences based on the methodology described in this study. However, the application of this method is limited when non-structured wiring data is used as the data source. In such cases, non-structured data should first be converted into structured data using other techniques before applying the method for automatic wiring list manual generation.

In conclusion, this method significantly improves the efficiency of compiling civil aircraft wiring list technical manuals and enhances content accuracy. Its feasibility and practicality have been fully validated, demonstrating excellent application prospects in the automatic generation of wiring lists for civil aircraft. Furthermore, it holds potential for broader application in other fields, such as the automotive and maritime industries.

REFERENCES

- [1] International Civil Aviation Organization. Summary of discussions, cross-border transferability (XBT)[R]. Montreal: ICAO, 2016.
- [2] Civil Aviation Administration of China. Aircraft continuous airworthiness documentation : AC-91-11 [S]. Beijing : Civil Aviation Administration of China, 2014.
- [3] Wan J Z. Study of Development IETM Based on S1000D Specification [J]. Applied Mechanics and Materials,2012,1923(190-191):249-252.
- [4] Niu Qingping. Research and implementation of interactive electronic technical manual for EMU air-conditioning system[D]. Lanzhou Jiaotong University, 2023.
- [5] Xuan Zhenyuan, Zhan Kexin, Yang Xin, Li Guoshun. Research on IETM Management Platform for EMUs Based on S1000D[J]. Railway Computer Application, 2022, 31 (03): 42-47.
- [6] Wei Yanfeng, Qiao Linjun. Research on Structuring Method of Aircraft Technical Publication Based on S1000D Standard[J]. Progress in Aeronautical Engineering, 2021, 12 (05): 116-122.
- [7] HE Chao. Introduction to the query method of Airbus AIRN@V line manual[J]. Science & Technology Information, 2018, 16 (12): 62-63+65.
- [8] International specification for technical publications S1000D Issue 3.0. 2007-07-31.
- [9] Guo D, Du X. A Fault Isolation Method Based on S1000D Fault Data Module[J].IOP Conference Series: Materials Science and Engineering,2019,563(4):
- [10] Liu Yiming, Wang Xiaodong, Chen Jie. Construction of Aero Engine Technical Data System Based on S1000D Standard[J]. Aerodynamics, 2022, (05): 59-63.
- [11] Xu Buxiang, Jiang Qingxi, Wang Lina. Research on Compilation Method of Schematic Atlas of Aircraft System Based on S1000D[J]. Aviation Standardization and Quality, 2022, (03): 9-12.
- [12] Xu Juan, Xu Yanping, Yang Pengfei. Research on the Development of Structural Repair Manual for Domestic Civil Aircraft Based on S1000D[J]. Aviation Maintenance and Engineering, 2021, (07): 42-45.
- [13] Wang Haowen, Tang Honghao, Song Jia. Research on Application of Boeing 787 Data Module Based on S1000D Standard[J]. Aviation Maintenance and Engineering, 2023, (09): 30-31.
- [14] Wu Peng, Wang Chunyang, Wen Jingqian, Zou Liping. Research and Development of Digital Manufacturing System for Aircraft Wiring Harness[J]. Aeronautical Manufacturing Technology, 2017, (07): 67-71.
- [15] Gong Yan. Scheme of intelligent generation system for technical publication line data based on S1000D standard[A] Exploration, Innovation, Exchange: Proceedings of the 6th Youth Science and Technology Forum of the Chinese Society of Aeronautics and Astronautics (Volume II)[C]. Chinese Society of Aeronautics and Astronautics, Chinese Society of Aeronautics and Astronautics, 2014: